

E-Commerce Backend Project Documentation

Student Details

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1. Project Overview

This project is a backend e-commerce system built using FastAPI. It supports user management, product catalog, shopping cart, and order processing. The system is designed to be scalable, secure, and efficient for online shopping applications.

2. Features

- User registration and authentication (JWT-based login)
- Product management (add, update, delete, view products)
- Shopping cart functionality
- Order placement and tracking
- Secure password hashing
- RESTful API design using FastAPI

3. Tech Stack

- Backend: FastAPI
- Database: SQLite / MySQL / PostgreSQL (optional)
- ORM: SQLAlchemy
- Authentication: JWT (python-jose)
- Password Hashing: bcrypt (passlib)

4. Project Structure

```
app/  
■■■ main.py  
■■■ models.py  
■■■ schemas.py  
■■■ database.py  
■■■ routes/  
■ ■■■ auth.py  
■ ■■■ products.py  
■ ■■■ cart.py  
■ ■■■ orders.py  
■■■ utils/  
■■■ security.py
```

5. Requirements

fastapi
uvicorn
sqlalchemy
pydantic
python-multipart
jinja2
passlib[bcrypt]
python-jose[cryptography]
python-dotenv

6. Conclusion

This project demonstrates a complete backend solution for an e-commerce platform. It provides a strong foundation for scalable web applications and can be extended with payment integration, advanced search, and deployment on cloud platforms.